Unit - II: Partial Differentiation

Tutorial Sheet – 2

1. If $u = x\text{^}\{log\ y\}\ where\ x^3 + y^3 + 3xy = 1, find\ \frac{du}{dx}$
2. **If** $u = f((y - x)/(xy), (z - x)/(xz)), show\ that: x^2 \frac{\partial u}{\partial x} + y^2 \frac{\partial u}{\partial y} + z^2 \frac{\partial u}{\partial z} = 0$
3. If $z = f(x, y) where\ x = e^u \cos v, y = e^u \sin v$ prove that

   $x \frac{\partial z}{\partial v} + y \frac{\partial z}{\partial u} = e^{2u} \frac{\partial z}{\partial y} (ii)\ (\partial z/\partial x)^2 + (\partial z/\partial y)^2 = [(\partial z/\partial u)^2 + (\partial z/\partial v)^2]$.

4. If $u = f(r, s, t)\ where\ r = y/x, s = z/y, t = x/z$, prove that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = 0$
5. If $u = sin^{-1}(x - y) where\ x = 3t, y = 4t^3, show\ that \frac{du}{dt} = \frac{3}{\sqrt{1 - t^2}}$
6. If $u = f(x^2 + 2yz, y^2 + 2zx)$, then show that

   $(y^2 - zx) \frac{\partial u}{\partial x} + (x^2 - yz) \frac{\partial u}{\partial y} + (z^2 - xy) \frac{\partial u}{\partial z} = 0$
7. If $u = f(2x - 3y, 3y - 4z, 4z - 2x)$,then show that $\frac{1}{2}\frac{\partial u}{\partial x} + \frac{1}{3}\frac{\partial u}{\partial y} + \frac{1}{4}\frac{\partial u}{\partial z} = 0$
8. If $u = f\left(\frac{y-x}{xy}, \frac{z-x}{xz}\right)$ then show that $x^2 u_x + y^2 u_y + z^2 u_z = 0$.
9. If $u = f(x, y)\ where\ x = r\ cos\ \theta, y = r\ sin\ \theta$, then show that

   $(u_x)^2 + (u_y)^2 = (u_r)^2 + \frac{1}{r^2} (u_\theta)^2$